Tobias Géron

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Research interests: Time-Domain Astronomy, Stellar Astrophysics, Galaxy Evolution, Machine Learning, Software Development, Citizen Science

Work Experience

Nov 2023 – Current **Postdoctoral Fellow | University of Toronto | Toronto, Canada**
I am a Canadian Rubin Postdoctoral Fellow at the Dunlap Institute of Astronomy and Astrophysics, University of Toronto. In addition to research, I develop software tools to facilitate science within the Transients and Variable Stars Collaboration of Rubin LSST.

Education

Sep 2019 – May 2023 **PhD in Astrophysics | University of Oxford | Oxford, United Kingdom**
Thesis title: *Strongly and weakly barred galaxies in Galaxy Zoo*
Advisors: Prof. Chris Lintott and Dr. Rebecca Smethurst

Sep 2018 – Sep 2019 **MSc in Astrophysics | University of St Andrews | St Andrews, United Kingdom**
Master thesis: *Mapping dark matter in MaNGA post-starburst galaxies*
Advisors: Dr. Anne-Marie Weijmans and Dr. Vivienne Wild

Dec 2016 – Jul 2018 **MSc in Bioscience Engineering | Ghent University | Ghent, Belgium**
Master thesis: *Natural killer cell-mediated delivery of nanoparticles as a new revolutionary approach to cancer immunotherapy*
Advisors: Prof. Koen Raemdonck and Prof. Daisy Vanrompay

Aug 2016 – Dec 2016 **Exchange Student | Yonsei University | Seoul, South Korea**

Sep 2013 – Aug 2016 **BSc in Bioscience Engineering | University of Antwerp | Antwerp, Belgium**
Bachelor Thesis: *Optimization of high-content microscopy methods to research cell-cycle dependent distribution of protein complexes in laminopathy models*
Advisor: Prof. Winnok De Vos

Summary of Research Activities

Publications	6 first author; 183 total citations 23 co-author; 2,008 total citations	Talks and presentations	7 invited talks 27 contributed talks
Teaching and Mentoring	2 courses taught 11 students supervised		

Leadership & Service

Jun 2026 – Current	Technical Lead | Galaxy Zoo
Apr 2026 – Current	Leadership team for LSST-DA Catalyst Symposium 2026 | LSST Discovery Alliance
Apr 2026 – Current	LOC lead for LSST-DA Catalyst Symposium 2026 | LSST Discovery Alliance
Apr 2026 – Current	SOC for LSST-DA Catalyst Symposium 2026 | LSST Discovery Alliance
Feb 2026 – Current	Organizer of Catalyst Postdoc Alliance Mentoring Program
Feb 2026 – Current	Founder and Organizer of the Catalyst Postdoc Alliance Talk Series
Feb 2026 – Current	Founder and Organizer of the Catalyst Postdoc Alliance Co-working Sessions
Jan 2026 – Current	SOC for Special Session on Rubin LSST | European Astronomical Society Meeting

Sep 2025 – Current	Catalyst Postdoctoral Alliance Lead | LSST Discovery Alliance I serve as the inaugural Catalyst Postdoctoral Alliance Lead for the Rubin LSST Discovery Alliance. I head a team of 40 postdoctoral researchers at institutions around the world that are all involved with the Rubin Observatory. I help to facilitate collaborations across institutions and foster community engagement among postdocs. More information about my role can be found here: `https://lsstdiscoveryalliance.org/geron-leads-catalyst-postdoc-alliance`.
Sep 2025 – Current	Postdoc Representative of the Training and Mentoring Committee | University of Toronto
Nov 2023 – Current	Peer reviewer for ApJ and MNRAS
Jul 2026	Session Chair for "Transient Science" | Rubin Community Workshop 2026
May 2026	Moderator for the "Hogg Visitorship Panel Discussion" | University of Toronto
2026	Proposal reviewer for NASA Panel | NASA
2026	Proposal reviewer for LSST-DA Panel | LSST Discovery Alliance
Sep 2025 – May 2026	Hogg Distinguished Visitorship Committee | University of Toronto
Oct 2025	Panel Moderator for "STEM Careers Outside of a University Setting" | University of Toronto
Sep 2025	Panel Moderator for "Making a Career Shift from Astronomy to a Data-Related Field" | University of Toronto
Sep 2025 – Nov 2025	SOC for LSST-DA Catalyst Symposium 2025 | LSST Discovery Alliance
Feb 2025 – Mar 2025	Dunlap Instrumentation Summer School Admissions Committee | University of Toronto

Awards & Grants

May 2026 – Current	Dunlap Institute Conference & Workshop Fund Award (C$ 5,100)
Sep 2025 – Current	LSST-DA Catalyst Postdoc Alliance Lead Fellowship Award ($ 41,921)
Jan 2025	Haverford's Early Career Scholarship ($ 1,300)
Sep 2019 – May 2023	Oxford Physics Endowment for Graduates Scholarship (£ 54,737)
Sep 2019 – May 2023	Saven European Scholarship (£ 28,235)
Sep 2019	Dean's List Award for Academic Excellence (MSc in Astrophysics)
July 2018	Graduated *Magna Cum Laude* (MSc in Bioscience Engineering)
Aug 2016	Graduated *Cum Laude* (BSc in Bioscience Engineering)
Aug 2016 – Dec 2016	ASEM-DUO Fellowship (€ 3,687.5)

Collaborations & Professional Memberships

Mar 2026 – Current	Rubin Alert Explorers | Principal Investigator
Sep 2025 – Current	Bar Brigade on the Experiment Platform | Co-Investigator
Sep 2025 – Current	Catalyst Postdoc Alliance (CPA) | Inaugural Lead
Sep 2025 – Current	Canadian Data-Intensive Astrophysics PLatform (CanDIAPL)
Sep 2025 – Current	LSST Discovery Alliance (LSST-DA)
Mar 2025 – Current	Canadian Astronomical Society - Société Canadienne d'Astronomie (CASCA)
Mar 2024 – Current	American Astronomical Society (AAS)
Nov 2023 – Current	Transient and Variable Star Science Collaboration (TVS SC) of Rubin LSST
Nov 2023 – Current	Software Task Force of TVS SC
Sep 2019 – Current	Galaxy Zoo | Core Member (since Sep 2019) and Technical Lead (since Jun 2026)
Sep 2019 – May 2023	European Astronomical Society (EAS)
Sep 2019 – May 2023	Royal Astronomical Society (RAS)
Sep 2019 – May 2023	Mapping Nearby Galaxies at APO (MaNGA)
Sep 2019 – May 2023	Sloan Digital Sky Survey (SDSS)

Teaching Experience

Jul 2026	Research Software Engineering Mini-Course | Co-Instructor | Menlo Park, California, USA I was one of the co-instructors for this mini-course organized at the Rubin Community Workshop 2026 at SLAC. I taught the section "Automatically Testing Software".

Jul 2025 – Aug 2025 Stars and Galaxies (AST201) | Head Instructor | University of Toronto | Toronto, Canada
This was an undergraduate course aimed at non-science majors, with 196 students enrolled. I led a team of six teaching assistants. I scored particularly well in my teaching evaluations for quality of instruction and student enthusiasm for the course. My overall score (4.7/5) was notably higher than the divisional average (4.2/5). Full course evaluations are available upon request.

Jan 2025 Programming in Astronomy (AST104) | Guest Lecture | Haverford College | Haverford, Pennsylvania, USA

Jul 2024 – Aug 2024 Stars and Galaxies (AST201) | Co-Instructor | University of Toronto | Toronto, Canada

Examples of student feedback can be found here: `https://tobiasgeron.github.io/teaching_overview.html`. Full teaching evaluations are available upon request.

Supervising & Mentoring Experience

Sep 2025 – Current Brian Aziz | Primary Supervisor
Undergraduate Research Project | University of Toronto

Sep 2025 – Current Keir Maloney | Primary Supervisor
Undergraduate Research Project | University of Toronto

Jan 2025 – Current Nasser Mohammed | PhD Mentor
PhD Mentoring Program | University of Toronto

May 2026 – Jul 2026 Alexis Li | Primary Supervisor
Summer Undergraduate Research Project | University of Toronto

May 2026 – Aug 2026 Caroline Teel | Primary Supervisor
Undergraduate Research Project | University of Toronto

Jul 2025 – Aug 2025 Jadaya Henry-Richards | Primary Supervisor
High School Internship through Visions of Science | University of Toronto

Jul 2025 – Aug 2025 Fadumo Abokor | Primary Supervisor
High School Internship through Visions of Science | University of Toronto

May 2025 – Aug 2026 Maëlle Magnan | Primary Supervisor
Summer Undergraduate Research Project | University of Toronto

Jul 2024 – Aug 2025 Tessa Pearlstein | Co-supervisor
Senior Thesis Project | Haverford College

Sep 2024 – Apr 2025 Aaryan Thusoo | Primary Supervisor
Undergraduate Research Project | University of Toronto

Feb 2024 – Sep 2024 Petra Mengistu | Co-supervisor
Senior Thesis Project | Haverford College

Observing Experience

23 Sep 2022 – 27 Sep 2022 Isaac Newton Telescope | Roque de los Muchachos Observatory | La Palma, Spain
25 Mar 2019 – 31 Mar 2019 Teide Observatory | Tenerife, Spain
Sep 2018 – Sep 2019 St Andrews University Observatory | St Andrews, United Kingdom

Software Packages

MultiStarFitting: Includes routines to rapidly fit single or binary stellar template models to data using any custom template library and any set of filters. Currently in development.

flexAE: A flexible auto-encoder framework that seamlessly incorporates multiple state-of-the-art AE components. Used in Géron et al. (submitted) to identify unresolved binaries in data from Rubin DP1.
Code available here: `https://github.com/tobiasgeron/flexAE`

DETECT: Quantify detection thresholds and upper limits in any image from Rubin LSST by performing a series of source injection, image subtraction, and forced photometry on a case-by-case basis. This code was introduced

and used in Géron et al. (2026a) to look for pre-SN variability in data from Rubin DP1.
Code available here: `https://github.com/tobiasgeron/detect`

BarPy: A collection of various techniques commonly used in the astronomical literature on images of barred galaxies. Currently includes ellipse fitting techniques. Fourier-based techniques coming soon.
Code available here: `https://github.com/tobiasgeron/barpy`

Tremaine-Weinberg: Calculate the bar pattern speed, corotation radius and $\mathcal{R}$ of barred galaxies using the Tremaine-Weinberg method. This was originally developed to work with data from the MaNGA survey, but now works with data from any IFU. This code was used in Géron et al. (2023), Géron et al. (2024), Qiu et al. (2025), Pearlstein et al. (2025), and Mengistu et al. (2026).
Code available here: `https://github.com/tobiasgeron/Tremaine_Weinberg`

Coordinate_matching: A fast pipeline to facilitate cross-matching different astronomical catalogues that automatically removes duplicates and can match recursively to ensure the maximum number of reliable matches.
Code available here: `https://github.com/tobiasgeron/coordinate_matching`

The code for all my packages is open-source and publicly available on GitHub: `https://github.com/tobiasgeron`.

Talks

29 Jul 2026	Rubin Community Workshop 2026 | SLAC | Menlo Park, California, USA Talk: Looking for pre-SN variability in Rubin LSST
19 Jun 2026	Astro on Tap | University of Toronto | Toronto, Ontario, Canada Invited talk: How Bars Kill Galaxies
11 Jun 2026	MAMSIE Group Meeting | KU Leuven | Leuven, Belgium Talk: Finding binary systems in Rubin LSST: autoencoders + template fitting
9 Jun 2026	Stars Group Meeting | KU Leuven | Leuven, Belgium Talk: Looking for pre-SN variability with Rubin LSST
7 Apr 2026	Teaching the Universe 2026 | University of Toronto | Toronto, Ontario, Canada Invited talk: What is the Rubin Observatory? And how do I use it to study dying stars?
7 Jan 2026	American Astronomical Society Winter Meeting 2026 | Phoenix, Arizona, USA Talk: DETECT: A pipeline to quantify detection thresholds in Rubin
20 Nov 2025	LSST-DA Member Representatives Meeting | Tucson, Arizona, USA Invited talk: DETECT: A pipeline to quantify detection thresholds in Rubin
16 Oct 2025	CanDIAPL Collaboration Meeting | University of Waterloo | Waterloo, Ontario, Canada Talk: DETECT: A pipeline to quantify detection thresholds in Rubin
16 Oct 2025	Canadian Rubin Fellows Meeting | University of Waterloo | Waterloo, Ontario, Canada Talk: Looking for pre-SN variability: first in ATLAS, now in Rubin
14 Oct 2025	GEESE-ON | University of Waterloo | Waterloo, Ontario, Canada Talk: Galaxy Zoo with JWST
3 Jun 2025	Canadian Astronomical Society Annual Meeting 2025 | Saint Mary's University | Halifax, Nova Scotia, Canada Talk: Galaxy Zoo CEERS: Bar fractions up to $z \sim 4$
20 May 2025	Shaping the Future of Time-Domain Astronomy with LSST | Centro Brasileiro de Pesquisas Físicas | Rio de Janeiro, Brazil Talk: Quantifying detection thresholds using Rubin pipelines to constrain pre-SN variability
28 Mar 2025	Astroseminar | Carnegie Mellon University | Pittsburgh, Pennsylvania, USA Invited talk: The effects of bar kinematics and bar strength on galaxy evolution
25 Mar 2025	TVS Software Workshop | Virtual Talk: Constraining pre-SN variability by quantifying detection thresholds using the Rubin pipelines
19 Mar 2025	Galaxy Zoo Team Meeting | Bern, Switzerland Talk: GZ CEERS: Bar fractions up to $z \sim 4.0$ (and the challenges along the way. . .)

19 Feb 2025 Astroseminar | University of Waterloo | Waterloo, Ontario, Canada
Invited talk: The effects of bar kinematics and bar strength on galaxy evolution

30 Jan 2025 Public Talk | Haverford College | Haverford, Pennsylvania, USA
Invited talk: GZ CEERS: Bar fractions up to $z \sim 4.0$

28 Jan 2025 Galaxy Group Talk | Haverford College | Haverford, Pennsylvania, USA
Talk: The effects of bar kinematics and bar strength on galaxy evolution

23 Jan 2025 AstroCoffee | Johns Hopkins University | Baltimore, Maryland, USA
Talk: The effects of bar strength and kinematics on galaxy evolution: Slow strong bars affect their hosts the most

23 Jan 2025 GREENS Talk | Space Telescope Science Institute | Baltimore, Maryland, USA
Talk: GZ CEERS: Bar fractions up to $z \sim 4.0$

21 Jan 2025 Galaxies and AGN Seminar | Space Telescope Science Institute | Baltimore, Maryland, USA
Talk: The effects of bar kinematics and bar strength on galaxy evolution

14 Jan 2025 American Astronomical Society Winter Meeting 2025 | National Harbor, Maryland, USA
Talk: GZ CEERS: Bar fractions up to $z \sim 4.0$

6 Jan 2025 CEERS Telecon | Virtual
Talk: GZ CEERS: Bar fractions up to $z \sim 4.0$

10 Sep 2024 TASTY Talk | University of Toronto | Toronto, Canada
Talk: The impact of bar kinematics on star formation in strongly and weakly barred galaxies

13 Jun 2024 American Astronomical Society Meeting 2024 | Madison, Wisconsin, USA
Talk: The kinematics of strong and weak bars: strong and slow bars affect their hosts the most

3 Apr 2024 Galaxy Zoo Team Meeting | Bern, Switzerland
Talk: Bars in JWST

22 Feb 2024 Galaxy Zoo Telecon | Virtual
Talk: The effects of bar strength and kinematics on galaxy evolution

16 Feb 2024 FOEs Meeting | University of Toronto | Toronto, Canada
Talk: Introduction to Rubin LSST

6 Mar 2023 Lancaster Seminar | Lancaster University | Lancaster, United Kingdom
Talk: Kinematics of strongly and weakly barred galaxies

9 Aug 2022 International Astronomical Union General Assembly 2022 | Busan, South Korea
E-talk: Stronger bars facilitate quenching in star-forming galaxies

22 Nov 2021 MaNGA SciCon Seminar | Virtual
Talk: Stronger bars facilitate quenching in star-forming galaxies

9 Nov 2021 St Andrews Lunch Seminar | University of St Andrews | Virtual
Invited talk: Stronger bars facilitate quenching in star-forming galaxies

20 Nov 2020 Galaxy Zoo Telecon | Virtual
Talk: Galaxy Zoo: only the strongest bars contribute to quenching in star-forming galaxies

17 Nov 2020 Oxford Hitchhikers Seminar | University of Oxford | Virtual
Talk: Bars in Galaxy Zoo (and Oxford)

Publications

First author publications

Géron, T., et al. submitted: *Unresolved Binary Systems in the Rubin Era I: An Autoencoder Framework for Binary Identification Applied to 47 Tucanae*

Géron, T., et al. 2026, ApJ, 1007, 47: *The Role of Baryonic and Dark Matter in Bar Kinematics*

Géron, T., et al. 2026, AJ, 172, 45: *DETECT: A Pipeline to Quantify Detection Thresholds in Rubin for Nearby Targets Embedded in Bright Host Galaxies*

Géron, T., et al. 2025, ApJ, 987, 74: *GZ CEERS: Bar fractions up to* $z \sim 4.0$

Géron, T., et al. 2024, ApJ, 973, 129: *The Effects of Bar Strength and Kinematics on Galaxy Evolution: Slow Strong Bars Affect Their Hosts the Most*

Géron, T., et al. 2023, MNRAS, 521, 1775: *Galaxy Zoo: Kinematics of strongly and weakly barred galaxies*

Géron, T., et al. 2021, MNRAS, 507, 4389: *Galaxy Zoo: stronger bars facilitate quenching in star-forming galaxies*

Second or third author publications

McClure, R. L., **Géron, T.**, D'Onghia, E., et al. submitted: *Secular Attrition of Classical Bulges by Stellar Bars*

Smethurst, R. J., Simmons, B. D., **Géron, T.**, et al. 2025, MNRAS, 539, 1359: *Galaxy Zoo JWST: up to 75 per cent of discs are featureless at* $3 < z < 7$

Walmsley, M., **Géron, T.**, Kruk, S., et al. 2023, MNRAS, 526, 4768: *Galaxy Zoo DESI: Detailed Morphology Classifications for 8.7m Galaxies in the DESI Legacy Imaging Surveys*

Walmsley, M., Smethurst, R. J., **Géron, T.**, et al. 2022, MNRAS, 509, 3966: *Galaxy Zoo DECaLS: Detailed visual morphology measurements from volunteers and deep learning for 314 000 galaxies*

Student-led publications

Aziz, B., **Géron, T.**, Garland, I. L., et al. submitted: *Bar Kinematics in Paired and Isolated Galaxies: Slow Bars Are More Likely Found in Paired Galaxies*

Mengistu, P., Masters, K. L., **Géron, T.**, et al. 2026, MNRAS, 548, 561: *The effects of bar strength and kinematics on galaxy evolution II: The global and local impacts of slow-strong bars*

Powley, J. M., Smethurst, R. J., Lintott, C. L., **Géron, T.** 2026, MNRAS, 548, 747: *Introducing* ΔV_{*-g}*: a new universal kinematic disturbance parameter*

Puczek, N., **Géron, T.**, Smethurst, R. J., et al. 2026, MNRAS, 547, 175: *Bars in low-density environments rotate faster than bars in dense regions*

Magnan, M., **Géron, T.**, Garland, I. L., et al. 2025, RNAAS, 9, 341: *Strong Bars, Strong Inflow: The Effect of Bar Strength on Gas Inflow*

Pearlstein, T., Masters, K. L., **Géron, T.** 2025, RNAAS, 9, 236: *Measuring Bar and Spiral Arm Pattern Speeds in MaNGA Barred Spiral Galaxies*

Co-author publications

de Soto, K., et al. submitted: *AT 2024ahzi: A Type IIP Supernova Discovered by the LSST Commissioning Camera*

de Ugarte Postigo, A., et al. submitted: *HI and CO spectroscopy of the unusual host of GRB 171205A: A grand design spiral galaxy with a distorted HI field*

Hutchinson-Smith, T., et al. submitted: *Galaxy Zoo Bar Lengths: A Catalogue of Measurements from Hubble Space Telescope Images and the Evolution of Galactic Bar Structure at* $z < 1$

Popp, J. J., et al. submitted: *The fraction of clumpy star-forming galaxies in nearby galaxies from CLAUDS and HSC-SSPP*

Popp, J. J., et al. submitted: *Star-forming clump detection in nearby galaxies using Faster R-CNN and ugrizy imaging data from CLAUDS and HSC-SSP*

Spindler, A. et al. submitted: *Deep Learning Segmentation of Spiral Arms and Bars for 600,000 Galaxies in DESI*

Walmsley, M., et al. submitted: *Galaxy Zoo Evo: 1 million human-annotated images of galaxies*

Walmsley, M., et al. submitted: *Scaling Laws for Galaxy Images*

Wise, E., et al. submitted: *Exploring the Relationship Between Bars, Star Formation Activity, and Host Galaxy Properties from* $z \sim 0$ *to* $z \sim 2$

Garland, I. L., et al. 2026, A&A, 709, A48: *The complex relationships between AGN, bars and bulges*

Huertas-Company, M., et al. 2026, A&A, 711, A11: *Euclid Quick Data Release (Q1): XI. A first look at the fraction of bars in massive galaxies at* $z < 1$

Walmsley, M., et al. 2026, A&A, 711, A9: *Euclid Quick Data Release (Q1): IX. First visual morphology catalogue*

Fahey, M. J., et al. 2025, MNRAS, 537, 3511: *Structural decomposition of merger-free galaxies hosting luminous AGNs*

Huertas-Company, M., et al. 2025, A&A, 704, A94: *COSMOS-Web: The emergence of the Hubble Sequence*

O'Ryan, D., et al. 2025, MNRAS, 539, 2967: *Time-scales for the effects of interactions on galaxy properties and SMBH growth*

Thöne, C. C., et al. 2024, A&A, 690, A66: *The host of GRB 171205A in 3D: A resolved multiwavelength study of a rare grand-design spiral GRB host*

Garland, I. L., et al. 2023, MNRAS, 522, 221: *The most luminous, merger-free AGNs show only marginal correlation with bar presence*

O'Ryan, D., et al. 2023, ApJ, 948, 40: *Harnessing the Hubble Space Telescope Archives: A Catalog of 21,926 Interacting Galaxies*

Abdurro'uf, et al. 2022, ApJS, 259, 35: *The Seventeenth Data Release of the Sloan Digital Sky Surveys: Complete Release of MaNGA, MaStar, and APOGEE-2 Data*

Garma-Oehmichen, L., et al. 2022, MNRAS, 517, 5660: *SDSS IV MaNGA: Bar pattern speed in Milky Way Analogue galaxies*

Porter-Temple, R., et al. 2022, MNRAS, 515, 3875: *Galaxy And Mass Assembly: Galaxy Zoo spiral arms and star formation rates*

Smethurst, R. J., et al. 2022, MNRAS, 510, 4126: *Quantifying the poor purity and completeness of morphological samples selected by galaxy colour*

Smith, D., et al. 2022, MNRAS, 517, 4575: *Galaxy And Mass Assembly: galaxy morphology in the green valley, prominent rings, and looser spiral arms*

Walmsley, M., et al. 2022, MNRAS, 513, 1581: *Practical galaxy morphology tools from deep supervised representation learning*